Predicting Geographic Atrophy Risk Using Autofluorescence Imaging and Artificial Intelligence

Robert D. Slater, Apoorva Safai, Madeline Pflasterer-Jennerjohn, Jen Heathcote, Rachel E. Linderman, Pallavi Tiwari, Emily Chew, Amitha Domalpally

WISCONSIN READING CENTER
Department of Ophthalmology and Visual Sciences
UNIVERSITY OF WISCONSIN
SCHOOL OF MEDICINE AND PUBLIC HEALTH
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Background

- Geographic atrophy (GA) is the end stage of non-exudative AMD.
- Prevention strategies are essential to delay or prevent GA development.
- Identifying precursors of GA can help target individuals at risk and develop effective interventions.

Purpose

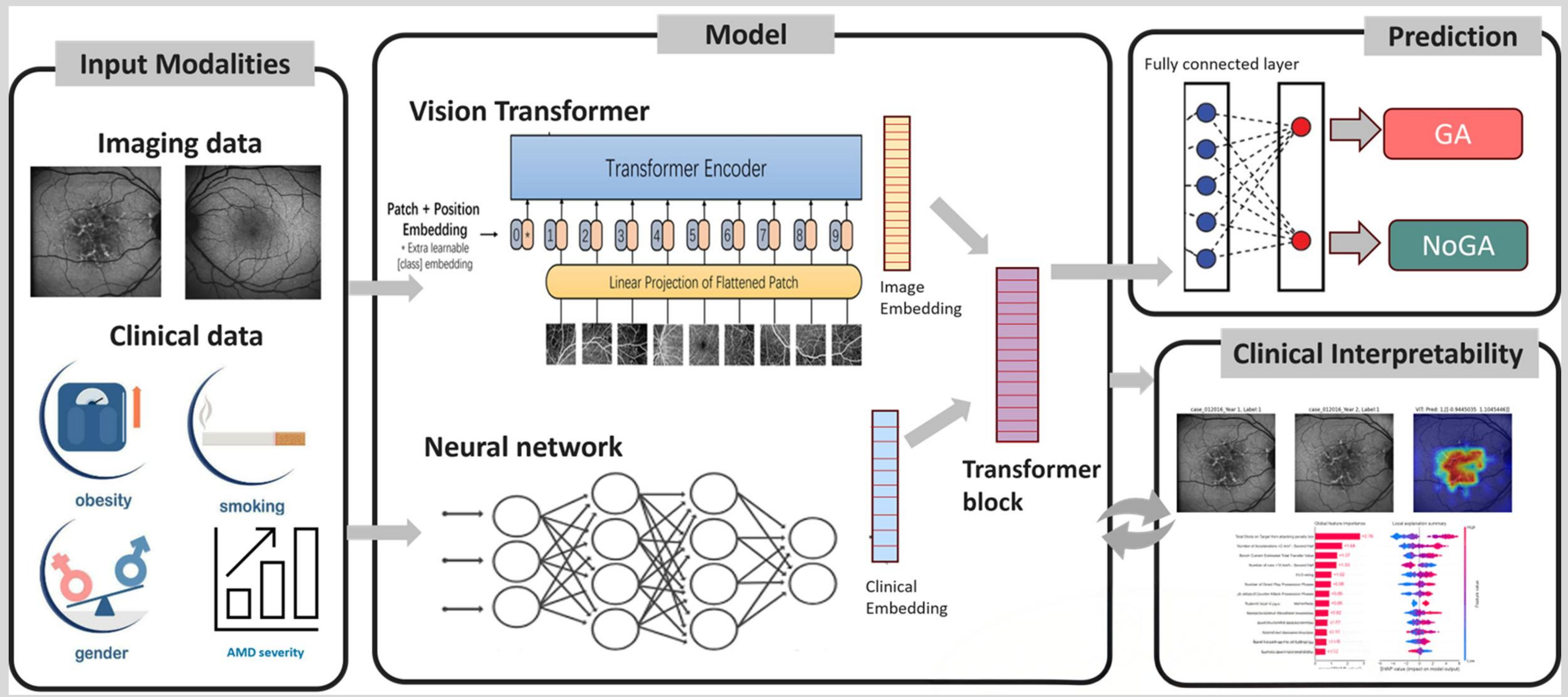
- To identify precursors of GA from autofluorescence(FAF) imaging using artificial intelligence (AI)
- To explore the utility of multimodal models combining demographic, clinical, and imaging data to enhance prediction accuracy

Methods

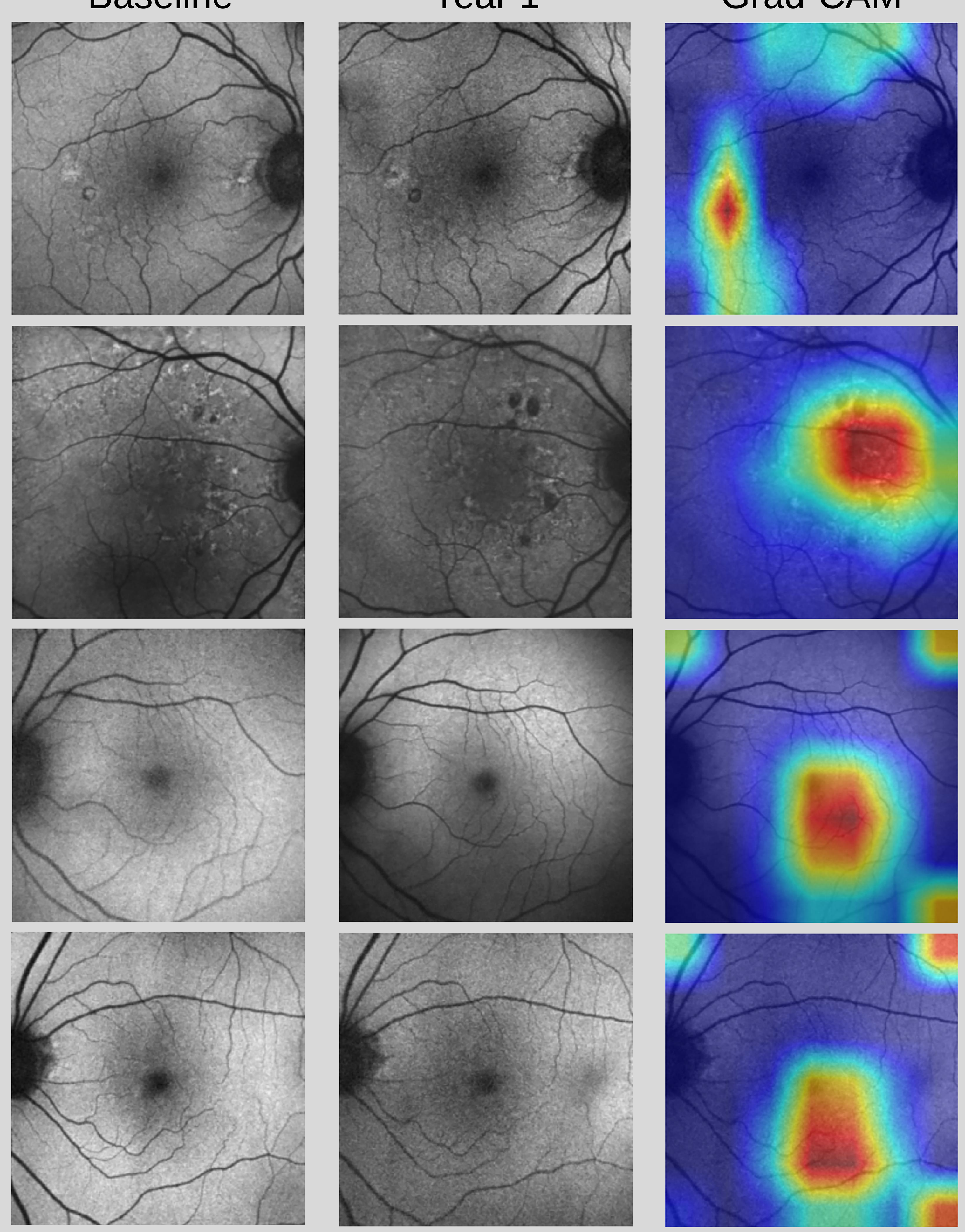
- AREDS2 FAF ancillary study
- Cases – eyes with incident GA
- Controls – eyes with no GA across 2 visits

FAF Imaging only	Clinical data only
Multimodal model Imaging + Clinical data	• Age, • Gender • smoking status • laterality • fellow eye status • RPD • AMD severity derived from Color photo

Image only : EfficientNet B1 with FAF images
Clinical model : 2-layer MLP neural network
Multimodal model : Transformer architecture



Baseline | Year 1 | Grad-CAM



Case 1: GA at year 1
- Age: 75
- Gender: M
- Smoking Status: former
- Laterality: RE
- Fellow Eye Status: NVAMD
- RPD: negative
- AMD Severity: 5

Case 2: GA at year 1
- Age: 71
- Gender: F
- Smoking Status: never
- Laterality: RE
- Fellow Eye Status : NVAMD
- RPD: positive
- AMD Severity: 7

Case 3: No GA at year 1
- Age: 73
- Gender: F
- Smoking Status: former
- Laterality: LE
- Fellow Eye Status: NVAMD
- RPD: negative
- AMD Severity: 4

Case 4: No GA at year 1
- Age: 71
- Gender: F
- Smoking Status: former
- Laterality: LE
- Fellow Eye Status: non-advanced
- RPD: negative
- AMD severity: 7

Results

	Cases (GA)	Controls (no GA)	Total
Training	71	132	203
Testing	11	44	55

Cross Validation Results

	Accuracy	Sensitivity	Specificity	AUROC	AUPRC
Image	95	94	95	97	95
Clinical	83	73	89	80	71
Multimodal	89	92	87	92	86

Image Clinical Multimodal

- The image model demonstrated highest accuracy followed by the multimodal model and clinical model
- Grad-CAM maps from FAF image model highlighted foveal activation where GA developed and a more diffuse activation when GA did not develop

Conclusions

- AI model trained on FAF imaging demonstrated superior predictive power for GA onset in iAMD patients compared to clinical factors alone.
- While FAF based AI model is promising, multimodal model may need additional investigation to identify the best combination of clinical and imaging models to further improve early prediction of GA.

References

1. Holmen et al. Precursors and Development of Geographic Atrophy with Autofluorescence Imaging: Age-Related Eye Disease Study 2 Report Number 18 Ophthalmology Retina Volume 3, Issue 9, September 2019, Pages 724-733

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